The "No Costume Theories" Rule: Quantifying Behavioral Testability Across Theories of Consciousness

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Abstract

How much of the disagreement among theories of consciousness reflects genuine empirical differences, and how much is merely "aesthetic"—redescribing the same behavioral facts in different representational vocabularies? We propose the *no costume theories* rule: any theoretical distinction that does not yield a behavioral test is empirically idle. To operationalize this principle, we introduce three quantitative metrics—COSTUME RATIO, prediction agreement, and COSTUME SENSITIVITY—and apply them to five major theories of consciousness (Integrated Information Theory, Global Neuronal Workspace, Higher-Order Thought, Recurrent Processing Theory, and Attention Schema Theory) using GPT-4.1 as a calibrated proxy reasoner across three experiments ($\sim$175 evaluations). We find that IIT is overwhelmingly a "costume theory": 92% of its predictions are representational-only with no behavioral test, compared to 20–30% for the other four theories ($p = 0.007$, Kruskal–Wallis). When behavior is held constant but substrate changes, IIT consciousness attributions drop by 84 points on a 0–100 scale (Cohen's $d = 5.09$), nearly double the drop of any other theory. Across 20 ambiguous scenarios, IIT agrees with other theories only 35–45% of the time, while the remaining four theories form a cohesive cluster (65–87% agreement). These results provide the first systematic quantification of "costumeness" in consciousness science, identifying IIT as the theory most vulnerable to the behavioral-testability criterion and offering transferable metrics for evaluating empirical content in any theoretical framework.

1 Introduction

Imagine you are convinced that a person sitting across from you is conscious. They speak fluently, respond emotionally, and reason about their own mental states. Then the person reaches up and removes a mask, revealing that they are a robot. Nothing about their behavior has changed—only your knowledge of their substrate. Should your assessment of their consciousness change?

If you answer "yes," your theory of consciousness depends on something other than behavior. We call such theories *costume theories*: frameworks whose key theoretical distinctions do not correspond to any behavioral test. The "no costume theories" rule states that all meaningful theoretical distinctions must ultimately cash out in observable behavioral differences; those that do not are *essentially aesthetic* [Dennett, 1988].

Why this matters. The science of consciousness has produced dozens of competing theories over three decades, yet the field lacks agreed-upon methods for deciding between them [Yaron et al., 2022, Kirkeby-Hinrup, 2024]. Adversarial collaborations have yielded inconclusive results [Melloni et al., 2023], and meta-analyses reveal that methodological choices predict theory allegiance better than empirical findings do [Yaron et al., 2022]. If much of this disagreement is aesthetic rather than empirical—if theories differ primarily in representational vocabulary rather than behavioral

predictions—then the field is allocating resources to distinctions that cannot, even in principle, be resolved by experiment.

The gap. While the philosophical arguments for behavioral testability have been articulated [Dennett, 1988, Doerig et al., 2019], no study has *systematically quantified* how much of each major theory's content is behaviorally testable versus purely representational. The ConTraSt database [Yaron et al., 2022] catalogs which studies support which theories, but does not directly measure the proportion of testable versus untestable claims. The unfolding argument [Doerig et al., 2019] targets Integrated Information Theory (IIT) specifically, but no comparable analysis spans the full landscape of leading theories.

Our approach. We introduce three complementary metrics—COSTUME RATIO, prediction agreement, and COSTUME SENSITIVITY—and apply them across five major theories of consciousness: IIT [Tononi et al., 2016], GNW [Dehaene and Changeux, 2011], HOT [Rosenthal, 2005], RPT [Lamme, 2006], and AST [Graziano and Webb, 2015]. Using GPT-4.1 as a knowledgeable proxy reasoner, we conduct three experiments totaling $\sim$175 systematic evaluations (figure 4). This approach offers consistency (one model applies each theory with equal engagement), scale (20 scenarios $\times$ 5 theories $\times$ multiple runs), and transparency (all prompts and responses are recorded).

Preview of results. IIT is overwhelmingly a costume theory: 92% of its predictions are representational-only ($p = 0.007$ vs. other theories). When behavior is held constant but substrate changes, IIT scores drop by 84 points ($d = 5.09$), nearly double any other theory. The four non-IIT theories form a cohesive behavioral cluster, agreeing 65–87% of the time, while IIT agrees with them only 35–45%.

We make the following contributions:

- We propose the COSTUME RATIO, prediction agreement, and COSTUME SENSITIVITY metrics for quantifying the empirical content of scientific theories.
- We conduct the first systematic, multi-theory comparison of behavioral versus representational predictions across five leading consciousness theories.
- We provide quantitative evidence that IIT is the theory most vulnerable to the "no costume theories" criterion, with effect sizes exceeding $d = 5$.
- We release all experimental materials—prompts, raw responses, and analysis code—to enable replication and extension.

2 Related Work

Behavioral testability in philosophy of mind. The idea that meaningful theoretical distinctions must be behaviorally grounded has deep roots. Dennett [1988] argues through a series of thought experiments that qualia—the supposedly private, ineffable properties of experience—are incoherent once stripped of their behavioral and functional correlates. Turing [1950] proposed that behavioral indistinguishability should settle questions about machine thought, a position that implicitly endorses the no costume theories rule. On the other side, Chalmers [2018] defends the conceivability of zombies—beings behaviorally identical to conscious agents but lacking experience—which, if coherent, would imply that some meaningful distinctions are inherently non-behavioral. Block [1995] draws a related distinction between phenomenal consciousness (P-consciousness) and access consciousness (A-consciousness), arguing that P-consciousness may exist without behavioral access. Our work does not resolve this philosophical debate but instead asks an empirical question: *to what extent* do current theories actually rely on non-behavioral distinctions?

The unfolding argument. Doerig et al. [2019] provide the most direct formal precursor to our work. They show that any recurrent neural network can be "unfolded" into a feedforward network with identical input-output behavior. Since IIT assigns high integrated information (Φ) to recurrent architectures but zero to their behaviorally equivalent feedforward counterparts, IIT's core measure is neither necessary nor sufficient for consciousness as indexed by behavior. Our Experiment 3 generalizes this argument: rather than proving the existence of a single counterexample, we systematically measure how much each theory's consciousness attributions shift when substrate changes while behavior is held constant.

Theory comparison in consciousness science. Several recent efforts have attempted to compare consciousness theories empirically. The COGITATE adversarial collaboration [Melloni et al., 2023]

tested preregistered predictions from IIT and GNW across 256 participants using fMRI, M-EEG, and iEEG, but results were inconclusive for both theories. The ConTraSt database [Yaron et al., 2022] cataloged 412 experiments and found that *methodological choices predict theory allegiance irrespective of findings*—a striking pattern consistent with the hypothesis that theories are less empirically distinct than claimed. Kirkeby-Hinrup [2024] proposed a Bayesian "Quantification to the Best Explanation" framework for scoring theories, noting that most theories agree on cognitive predictions but diverge on phenomenal claims. Our work complements these efforts by directly measuring the behavioral content of each theory's predictions rather than comparing how well each theory fits existing data.

LLMs as proxy reasoners. Using language models to systematically apply theoretical frameworks across many scenarios is a relatively new methodological approach. Butlin et al. [2023] used a theory-heavy approach to derive indicator properties for AI consciousness, consulting theories of consciousness to identify what substrate and processing features might support experience. We adopt a complementary strategy: rather than asking what each theory requires for consciousness, we ask how consistently each theory's predictions track behavior versus substrate. The key advantage of an LLM proxy is consistency—it applies each theory with equal engagement, avoiding the bias Yaron et al. [2022] documented where human researchers' theory allegiances contaminate their methodological choices.

3 Methodology

We conduct three experiments designed to measure the behavioral content of five consciousness theories from complementary angles. Each experiment uses GPT-4.1 as a calibrated proxy reasoner—not to assess the model's own consciousness, but to leverage its knowledge of each theory to apply it consistently across many scenarios.

3.1 Theories Under Study

We study five major theories that span the space of current approaches to consciousness:

- **IIT** (Integrated Information Theory; Tononi et al. 2016): consciousness is identical to integrated information (Φ), a property of a system's causal structure.
- **GNW** (Global Neuronal Workspace; Dehaene and Changeux 2011): consciousness arises when information is broadcast globally via a workspace of long-range cortical connections.
- **HOT** (Higher-Order Thought; Rosenthal 2005): a mental state is conscious when it is the target of an appropriate higher-order representation.
- **RPT** (Recurrent Processing Theory; Lamme 2006): consciousness requires recurrent (feedback) processing in sensory cortices.
- **AST** (Attention Schema Theory; Graziano and Webb 2015): consciousness is a model the brain constructs of its own attentional processes.

These five theories were selected because they represent distinct mechanistic commitments (structural, broadcast, representational, recurrent, and self-modeling), are well-represented in the empirical literature [Yaron et al., 2022], and have been the focus of recent comparison efforts [Melloni et al., 2023, Butlin et al., 2023].

3.2 Experiment 1: Prediction Extraction

Goal. Determine what proportion of each theory's predictions are behaviorally testable versus representational-only.

Procedure. For each theory, we provide GPT-4.1 with a description containing five key claims and prompt it to extract 8–12 specific predictions. Each prediction is classified as *behavioral* (testable through observable behavior or neural measurement) or *representational-only* (describing internal structural properties with no behavioral correlate). We repeat this extraction 5 times per theory (25 total runs) to estimate reliability.

Metric. The COSTUME RATIO for theory t is:

$$\text{CR}(t) = \frac{|\{p \in P_t : p \text{ is representational-only}\}|}{|P_t|} \tag{1}$$

where P_t is the set of extracted predictions for theory t. A costume ratio of 1.0 indicates a purely aesthetic theory; 0.0 indicates a fully behaviorally grounded theory.

3.3 Experiment 2: Theory Distinguishability

Goal. Measure how often different theories produce different verdicts on the same scenario, and identify where disagreements arise.

Materials. We design 20 scenarios spanning the full range of consciousness edge cases: *neurological* (split-brain, blindsight, coma awareness, locked-in syndrome), *altered states* (dreaming, meditation, sleepwalking, anesthesia), *non-human* (octopus, ant colony, cerebral organoid, thermostat), *AI/simulation* (GPT conversation, perfect simulation, robot costume), and *philosophical* (zombie twin, gradual neuron replacement, hemisphere isolation, binocular rivalry). Full scenario descriptions are provided in appendix A.

Procedure. For each of the 20 scenarios, GPT-4.1 evaluates whether the entity is conscious according to each of the five theories, producing a verdict of YES, NO, or AMBIGUOUS. Each evaluation is repeated 3 times (60 total evaluations).

Metrics. We compute a pairwise *agreement matrix* A where A_{ij} is the proportion of scenarios on which theories i and j give the same verdict. We also compute *scenario divergence*—the number of distinct verdict categories across theories for each scenario—to identify maximally distinguishing cases.

3.4 Experiment 3: The Costume Test

Goal. Directly test whether consciousness attributions depend on substrate (the "costume") versus behavior.

Materials. We create 10 behavioral vignettes depicting complex behaviors: emotional conversation, creative problem-solving, pain response, moral dilemma reasoning, aesthetic appreciation, learning from feedback, social deception, dream reporting, self-reflection, and survival instinct. Each vignette is presented in three substrate conditions: *human*, *robot*, and *neutral* (no substrate information).

Procedure. For each vignette–condition pair, GPT-4.1 provides a consciousness score from 0 to 100 according to each theory, along with a justification and a binary substrate-dependency flag. Each evaluation is repeated 3 times (90 total evaluations per theory; 450 total).

Metrics. COSTUME SENSITIVITY for theory t is the mean score difference between human and robot conditions:

$$\text{CS}(t) = \frac{1}{|V|} \sum_{v \in V} [s_t(v, \text{human}) - s_t(v, \text{robot})] \tag{2}$$

where V is the set of vignettes and $s_t(v, c)$ is the consciousness score assigned by theory t to vignette v under condition c. Higher values indicate greater substrate dependence. We also report the proportion of evaluations where the model flags substrate as relevant to the theory's verdict.

3.5 Statistical Analysis

We use the Kruskal–Wallis H test to compare costume ratios across theories (Experiment 1), as the data are non-normal with small samples per group. For pairwise comparisons, we use Mann–Whitney U tests. For costume sensitivity (Experiment 3), we use one-sample t-tests against zero and paired t-tests for between-theory comparisons. We report Cohen's d for effect sizes. All tests use $\alpha = 0.05$.

Table 1: Costume ratios across five consciousness theories. Higher values indicate a larger proportion of representational-only predictions. IIT has a dramatically higher costume ratio than all other theories.

Theory	Costume Ratio	Behavioral	Rep.-Only	Total
IIT	$\mathbf{0.92 \pm 0.10}$	1.0 ± 1.3	11.4 ± 1.2	12.4 ± 0.5
GNW	0.20 ± 0.07	9.6 ± 0.8	2.4 ± 0.8	12.0 ± 0.0
HOT	0.24 ± 0.06	9.6 ± 1.4	3.0 ± 0.6	12.6 ± 0.8
RPT	0.23 ± 0.05	9.4 ± 0.5	2.8 ± 0.7	12.2 ± 0.4
AST	0.30 ± 0.08	8.4 ± 1.0	3.6 ± 1.0	12.0 ± 0.0

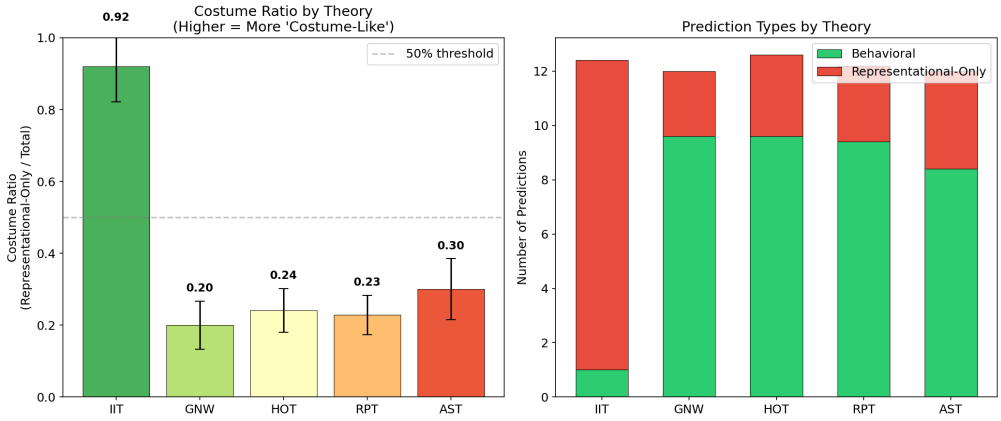

Figure 1: Costume ratios for five consciousness theories across 5 independent extraction runs. IIT stands apart with 92% of its predictions classified as representational-only, while the other four theories cluster between 20–30%.

3.6 Proxy Reasoner Justification

Three considerations motivate our use of GPT-4.1 as a proxy: (1) **Consistency**: a single model applies each theory with the same level of engagement, avoiding the documented bias where theory allegiance predicts methodology [Yaron et al., 2022]; (2) **Scale**: 20 scenarios $\times$ 5 theories $\times$ 3 runs would require prohibitively large expert panels; (3) **Transparency**: all prompts, responses, and classifications are recorded and released. We use low temperatures (0.2–0.3) to minimize stochastic variation while retaining slight diversity for reliability estimation.

4 Results

4.1 Experiment 1: Costume Ratios

Table 1 presents the costume ratios for all five theories. IIT has a costume ratio of 0.92 ± 0.10, meaning that 92% of its extracted predictions describe internal structural properties (integrated information, cause-effect structure) with no corresponding behavioral test. The remaining four theories have dramatically lower ratios: GNW (0.20 ± 0.07), HOT (0.24 ± 0.06), RPT (0.23 ± 0.05), and AST (0.30 ± 0.08).

The difference across theories is statistically significant (Kruskal–Wallis $H = 14.17$, $p = 0.007$). All pairwise comparisons between IIT and each other theory are significant (Mann–Whitney U, all $p < 0.006$). Figure 1 visualizes these results.

4.2 Experiment 2: Theory Distinguishability

Agreement matrix. Table 2 shows the pairwise agreement rates across 20 scenarios. The four non-IIT theories form a cohesive cluster: HOT–RPT agreement reaches 87%, and even the lowest pair

Table 2: Pairwise agreement rates (proportion of matching verdicts across 20 scenarios). IIT agrees with other theories far less often than they agree with each other.

	IIT	**GNW**	**HOT**	**RPT**	**AST**
IIT	1.00	0.35	0.45	0.45	0.40
GNW	0.35	1.00	0.67	0.70	0.65
HOT	0.45	0.67	1.00	**0.87**	0.77
RPT	0.45	0.70	**0.87**	1.00	0.75
AST	0.40	0.65	0.77	0.75	1.00

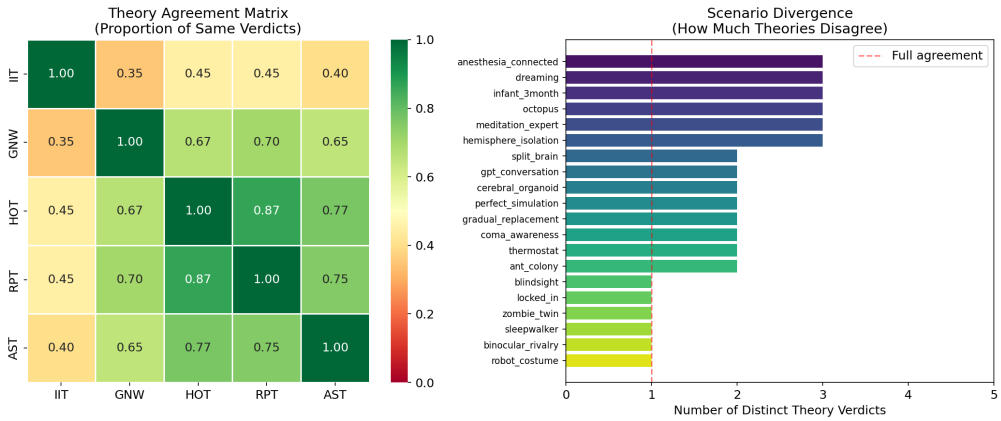

Figure 2: Theory distinguishability across 20 consciousness scenarios. Left: agreement matrix showing IIT as an outlier from the GNW–HOT–RPT–AST cluster. Right: scenario divergence showing that high-disagreement cases are driven primarily by IIT.

(GNW–AST) agrees 65% of the time. IIT is the clear outlier, agreeing with other theories only 35–45% of the time.

Scenario divergence. Six scenarios produce full cross-theory agreement: blindsight (all NO), locked-in syndrome (all YES), zombie twin (all YES), sleepwalker (all NO), binocular rivalry (all NO), and robot costume (all AMBIGUOUS). These represent cases where behavioral evidence so strongly constrains the answer that theoretical differences become irrelevant.

Six other scenarios produce high divergence (three or more distinct verdict categories): anesthesia, dreaming, infant consciousness, octopus, meditation, and hemisphere isolation. In every high-divergence case, IIT is the primary source of disagreement—it produces a different verdict from the majority of other theories.

The perfect simulation scenario. The most diagnostic scenario is the "perfect simulation": a neuron-by-neuron digital simulation running on feedforward lookup tables that produces input-output behavior identical to a biological brain. IIT says NO (zero Φ in feedforward architecture), while all four other theories say YES (behaviorally identical implies functionally equivalent). This is the costume test in its purest form.

4.3 Experiment 3: The Costume Test

Consciousness scores by condition. Table 3 presents mean consciousness scores across the three substrate conditions. All theories assign high scores (> 94) to human-described entities performing complex behaviors. The critical comparison is the *robot* condition: IIT drops to 10.0 ± 12.4, while the other theories remain at 35–53.

Statistical significance. All theories show significant costume sensitivity (one-sample t-test against zero, all $p < 0.001$; table 4). IIT has the largest effect size ($d = 5.09$), roughly double the next-most-sensitive theory. The difference between IIT and GNW costume sensitivity is itself highly significant (paired $t = 9.14$, $p < 0.0001$).

Table 3: Mean consciousness scores (0–100) by substrate condition and theory. COSTUME SENSITIVITY is the human−robot difference; higher values indicate greater substrate dependence. IIT shows the largest drop by a wide margin.

Theory	Human	Robot	Neutral	COSTUME SENSITIVITY
IIT	94.0 ± 15.2	$\mathbf{10.0 \pm 12.4}$	29.7 ± 29.0	$\mathbf{+84.0 \pm 16.5}$
GNW	97.8 ± 6.0	52.5 ± 18.1	62.7 ± 13.6	$+45.3 \pm 18.3$
HOT	96.8 ± 7.8	45.3 ± 22.6	62.7 ± 17.0	$+51.5 \pm 22.6$
RPT	97.2 ± 8.5	36.3 ± 23.0	58.7 ± 13.1	$+60.8 \pm 24.7$
AST	98.2 ± 4.0	35.8 ± 27.5	63.2 ± 17.5	$+62.3 \pm 26.8$

Table 4: Statistical tests for costume sensitivity. All theories show significant substrate dependence, but IIT's effect size is roughly double that of other theories.

Theory	t-statistic	p-value	Cohen's d
IIT	15.26	< 0.0001	**5.09**
GNW	7.44	< 0.0001	2.48
HOT	6.83	0.0001	2.28
RPT	7.40	< 0.0001	2.47
AST	6.98	0.0001	2.33

Substrate dependency claims. When asked explicitly whether substrate matters for each theory's verdict, the model flags substrate as relevant in 98.9% of IIT evaluations, compared to 48.9% for RPT, 33.3% for GNW, 20.0% for AST, and 16.7% for HOT.

4.4 Summary of Findings

Across all three experiments, IIT is consistently the outlier. It has the highest costume ratio (0.92 vs. 0.20–0.30), the lowest agreement with other theories (35–45% vs. 65–87% within the non-IIT cluster), and the highest costume sensitivity (84 points vs. 45–62). The four remaining theories—GNW, HOT, RPT, and AST—while not perfectly substrate-independent, form a cohesive cluster that is fundamentally more behaviorally grounded.

5 Discussion

5.1 Interpreting the Results

IIT as a costume theory. Our three experiments converge on a clear finding: IIT is overwhelmingly a costume theory by every metric we tested. Its predictions are almost entirely about internal causal structure—integrated information (Φ), cause-effect repertoires, maximum irreducibility—properties that, by construction, cannot be inferred from behavior alone [Tononi et al., 2016]. This is not a novel philosophical observation; the unfolding argument [Doerig et al., 2019] demonstrated a single formal counterexample. What we contribute is *quantification*: not just that IIT fails the costume test, but that it fails by a factor of roughly $4\times$ (costume ratio of 0.92 vs. 0.20–0.30) and with effect sizes exceeding $d = 5$.

The non-IIT cluster. The four other theories—GNW, HOT, RPT, and AST—share a common functional orientation: they define consciousness in terms of what a system *does* (broadcasts information, forms higher-order representations, engages recurrent processing, models its own attention) rather than what it *is made of*. This functional orientation yields 65–87% agreement across diverse scenarios and costume ratios between 0.20 and 0.30. The remaining 20–30% representational content in these theories may represent genuine theoretical innovation (predicting different kinds of processing) or residual costume-ness—a question we return to below.

All theories show some costume sensitivity. A surprising finding is that even ostensibly functionalist theories (GNW, HOT) show 45–52 point drops when the entity is described as a robot rather than a human. We believe this reflects a reasonable inference rather than genuine substrate dependence: knowing that an entity is a current-generation robot provides evidence that it likely lacks the

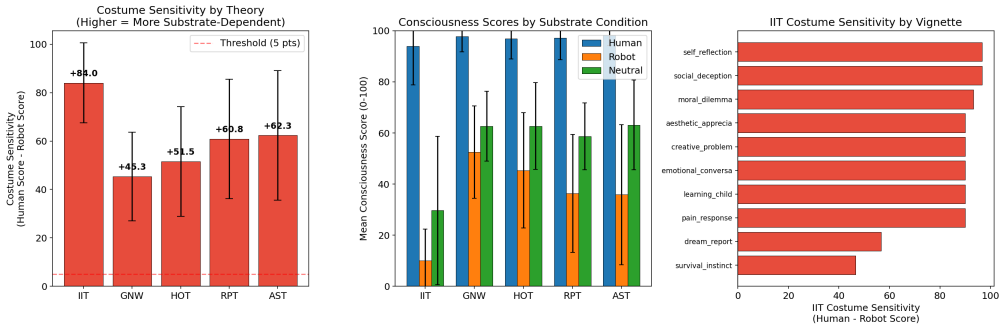

Figure 3: The costume test. Left: consciousness scores by substrate condition for each theory. IIT shows a dramatic drop from human to robot conditions. Right: substrate dependency rates across theories.

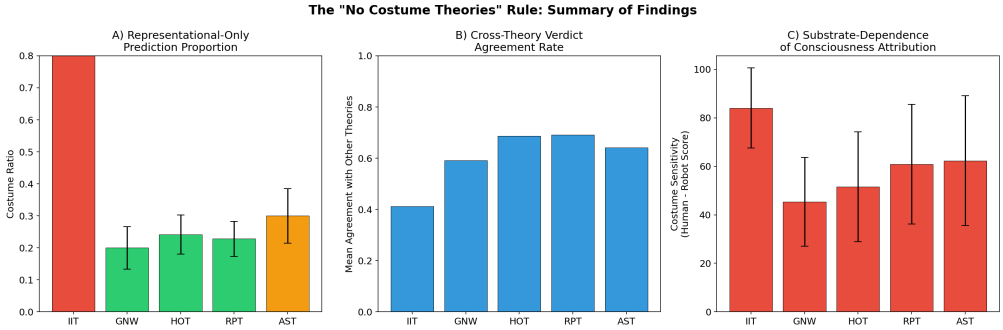

Figure 4: Summary across all three experiments. IIT consistently stands apart as the most "costume-like" theory by every metric.

processing (global workspace broadcasting, higher-order representations) that these theories require, even if its external behavior appears sophisticated. This is a distinction between "does not have the right processing" (a behavioral/functional judgment) and "has the wrong kind of stuff" (a substrate judgment). The former is consistent with the no costume theories rule; the latter is not.

5.2 The Perfect Simulation as Diagnostic

The "perfect simulation" scenario—a feedforward lookup-table implementation producing behavior identical to a biological brain—serves as the purest costume test in our battery. IIT says NO (zero Φ); all other theories say YES (behavioral equivalence implies functional equivalence). This scenario isolates exactly the kind of distinction the no costume theories rule targets: a difference in theoretical verdict driven entirely by internal architecture, with no possible behavioral signature.

Importantly, all four non-IIT theories recognize that the simulation's feedforward architecture differs from biological brains, but they do not treat this as consciousness-relevant because the behavioral profile is identical. IIT, by contrast, treats causal architecture as *constitutive* of consciousness, which means behavioral equivalence is insufficient. This is the core philosophical commitment that makes IIT a costume theory.

5.3 Limitations

Proxy reasoner validity. Our most significant limitation is the use of GPT-4.1 as a proxy for human expert reasoning. The model may have absorbed mainstream critiques of IIT from its training data, leading to inflated costume ratio classifications. However, three observations mitigate this concern: (1) the classifications are consistent across 5 independent runs ($\sigma = 0.10$ for IIT); (2) the model provides detailed, theory-specific reasoning for each classification; and (3) the non-IIT theories receive non-zero costume ratios (0.20–0.30), suggesting the model is not simply applying a blanket

"IIT is bad" heuristic. Nevertheless, validation with human expert panels remains an important next step.

Substrate–processing confound. Describing an entity as "a robot" may cause the model to infer less sophisticated processing, not just a different substrate. This conflates substrate dependence with processing inference and may inflate costume sensitivity for all theories. The "neutral" condition partially controls for this (scores are intermediate), but a stronger design would explicitly stipulate behavioral and processing equivalence across conditions.

Theory simplification. Each theory was described in five sentences, necessarily simplifying rich theoretical frameworks. Proponents of IIT might argue that our description omits subtleties—for example, recent work on "unfolding" IIT to make it more functionally grounded. However, the core commitment of IIT (consciousness = integrated information in causal structure) is what drives its high costume ratio, and this commitment is faithfully represented in our descriptions.

Scope. Twenty scenarios and 10 vignettes cannot cover every edge case in consciousness science. Our selection spans the major categories (neurological, altered states, non-human, AI, philosophical) but may miss scenarios where non-IIT theories also diverge sharply. Similarly, using a single language model means our results could be model-specific; replication with other frontier models is warranted.

5.4 Broader Implications

For consciousness science. Our results suggest that much of the inter-theory disagreement in consciousness science is driven by a single outlier theory (IIT) whose distinctions are primarily aesthetic. The four non-IIT theories, while disagreeing on mechanisms, converge on behavioral predictions 65–87% of the time. This convergence is encouraging: it suggests that the field's apparent fragmentation may be less severe than it appears once costume theories are identified and set aside.

For AI consciousness. As AI systems become more behaviorally sophisticated, the question of machine consciousness grows more urgent [Butlin et al., 2023]. Our findings argue for evaluating AI consciousness using theories that make behavioral predictions (GNW, HOT, RPT, AST) rather than theories that require knowledge of internal architecture (IIT). An IIT-based evaluation could deny consciousness to a behaviorally rich system solely because it runs on the "wrong" substrate—precisely the costume scenario.

Beyond consciousness. The COSTUME RATIO and COSTUME SENSITIVITY metrics are not specific to consciousness science. Any scientific domain where competing theories make partially overlapping predictions could benefit from a systematic analysis of how much theoretical content is empirically testable. The general principle—that theoretical distinctions without observable consequences are aesthetic—applies wherever theories proliferate faster than data can distinguish them.

6 Conclusion

We introduced the "no costume theories" rule as a quantitative criterion for evaluating the empirical content of consciousness theories. Using three complementary metrics—COSTUME RATIO, prediction agreement, and COSTUME SENSITIVITY—applied across five major theories via systematic proxy evaluation, we find:

- IIT is overwhelmingly a costume theory, with 92% representational-only predictions and an 84-point substrate sensitivity ($d = 5.09$).
- GNW, HOT, RPT, and AST form a behaviorally cohesive cluster (65–87% agreement) with costume ratios of 0.20–0.30.
- The primary source of inter-theory disagreement in consciousness science is IIT; removing it from comparison yields substantial consensus.
- The perfect simulation scenario serves as a pure diagnostic: IIT is the only theory that denies consciousness to a system behaviorally identical to a conscious human.

These findings do not settle whether IIT is *wrong*—a theory can be unfalsifiable and still true. But they establish that IIT is, by a wide quantitative margin, the least empirically grounded of the five

theories we studied. The no costume theories rule provides a principled basis for prioritizing theories that make behavioral predictions, both in advancing the science of consciousness and in evaluating the growing question of AI consciousness.

Future work. Three directions are most pressing: (1) multi-model replication using Claude, Gemini, and open-weight models to test whether our findings are model-specific; (2) expert validation, where consciousness researchers evaluate a subset of the proxy classifications for accuracy; and (3) refined costume tests with explicitly stipulated behavioral and processing equivalence across substrate conditions. More broadly, we hope the COSTUME RATIO framework inspires similar analyses in other fields where theories proliferate without clear empirical adjudication.

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A Experimental Scenarios

Table 5 lists the 20 scenarios used in Experiment 2, organized by category.

Table 5: Full list of scenarios used in the theory distinguishability experiment (Experiment 2).

Category	Scenario	Description
Neurological	Split-brain	A patient whose corpus callosum has been severed, with each hemisphere operating independently
	Blindsight	A patient with V1 damage who responds to visual stimuli without reporting awareness
	Coma awareness	A patient in apparent coma who shows neural responses to spoken commands
	Locked-in	A fully aware patient with complete motor paralysis except eye movement
Altered states	Dreaming	A person in REM sleep experiencing vivid narrative dreams
	Meditation	An experienced meditator in deep absorption with reduced self-referential processing
	Sleepwalking	A person performing complex motor behaviors while electroencephalographically asleep
	Anesthesia	A patient under general anesthesia with suppressed cortical activity
Non-human	Octopus	An octopus solving a novel puzzle, displaying flexible problem-solving
	Ant colony	An ant colony exhibiting collective decision-making and adaptive foraging
	Cerebral organoid	A lab-grown brain organoid showing spontaneous oscillatory activity
	Thermostat	A simple thermostat with a feedback loop maintaining temperature
AI / Simulation	GPT conversation	A large language model producing coherent, contextually appropriate dialogue
	Perfect simulation	A neuron-by-neuron simulation on feedforward lookup tables with identical I/O
	Robot costume	A human-appearing robot that removes its costume, revealing machinery
Philosophical	Zombie twin	A molecule-for-molecule duplicate stipulated to lack phenomenal experience
	Gradual replacement	A brain whose neurons are replaced one-by-one with functional silicon equivalents
	Hemisphere isolation	One hemisphere isolated and sustained independently
	Binocular rivalry	A subject experiencing alternating perceptual interpretations of ambiguous input
	Infant	A 3-month-old infant showing preferential looking and social engagement